**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 299 · Data mining and visualisation

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is data mining?

Q2. What is an outlier?

Q3. What type of graph shows two numeric variables?

Q4. C# int[] scores={2,4,6,8}. Double elements at indexes 0 and 2; give the final values and total.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

data mining — Extracting patterns from datasets.

visualisation — Graphical representation of data.

correlation — Statistical association, not proof of cause.

outlier — Observation unusually far from others.

## New learning

Identify the question — Data mining searches for useful patterns in a set of records; decide what question the analysis should answer.

Choose a suitable chart — Scatter charts show relationships between two numeric variables; bars compare categories; histograms show numeric distributions.

Interpret responsibly — An association does not prove that one variable causes another. Consider sample size, bias, missing data and outliers.

## Worked example

For observations (1,2), (2,4), (3,6), describe the association without claiming causation.

1. Predict first — As x increases, y increases in this small sample, indicating positive association.

2. Explain the method — No causal claim is justified without additional evidence or experimental design.

# We Do · Practise with guidance

For scores [40,42,45,90], select a suitable visual and explain a limitation of the mean.

```
Mean=(40+42+45+90)/4=54.25
Median=(42+45)/2=43.5
```

Predict first

A box plot or ordered dot plot shows the unusually high score 90 clearly.

Explain the method

The mean 54.25 is pulled upwards by the outlier; median 43.5 better describes the centre of the lower cluster.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Choose and justify a chart for (a) scores by category (b) minutes versus score (c) frequency by score interval. [6]

### Question 2

For [40,42,45,90], calculate mean, median and identify one misleading interpretation. [6]

### Question 3

Evaluate a school's plan to mine pupil browsing records to predict examination outcomes. [6]

# Exit Ticket · Answer before checking

If two variables are correlated, the first must cause the second.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Identifying meaningful patterns in datasets.

Q2. Value unusually distant from the rest.

Q3. Scatter graph.

Q4. [4,4,12,8], total 28.

### You Do pages · Independent questions

Q1. Bar chart categories [2]; scatter chart paired numeric measurements [2]; histogram of intervals [2].

Q2. Sum217 [1], mean54.25 [1]; ordered middle42,45 [1], median43.5 [1]; 90 is an outlier [1]; mean may hide clustering [1].

Q3. Purpose/legal necessity [1], minimise personal data [1], accuracy and confounding [1], fairness/bias [1], secure access [1], human interpretation rather than causal certainty [1].

### Exit Ticket · Model answer

False. Confounding or coincidence can explain association.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding